

CS 736: Medical Image Computing

(Instructor : Suyash P. Awate)

Mid-Semester Examination. Closed Book/Notes.

Duration: 2 hours

Roll Number: _____

Name: _____

When in doubt, make suitable assumptions, state them clearly, and proceed to solve the problem.
All the best!

Formulae for Reference

Poisson probability mass function: $P(x; \lambda) := \lambda^x \exp(-\lambda)/x!$ with mean = λ , variance = λ , and mode = $\text{floor}(\lambda)$.

Rician probability density function: $P(x; \nu, \sigma) := \frac{x}{\sigma^2} \exp(-\frac{x^2 + \nu^2}{2\sigma^2}) I_0(\frac{x\nu}{\sigma^2}), \forall x \geq 0$

Gamma probability mass function: $P(x; \alpha, \beta) := \eta(\alpha, \beta) x^{\alpha-1} \exp(-\beta x)$, where $\eta(\alpha, \beta)$ is a normalization constant that is a function of parameters α and β . In this case, mean = α/β , variance = α/β^2 , and mode = $(\alpha - 1)/\beta$.

Rayleigh probability density function: $P(x; \sigma) := \frac{x}{\sigma^2} \exp(-\frac{x^2}{2\sigma^2}), \forall x \geq 0$

Matrix/vector derivatives: $d(x^\top Ax) = x^\top (A + A^\top) dx$

1. [30 points] Consider a standard clustering problem with N data points $\{x_i \in \mathbb{R}^D\}_{i=1}^N$ in a D -dimensional Euclidean space to be clustered into K groups/clusters.

Let each group be represented by a mean $\mu_k \in \mathbb{R}^D$ as well as a unit-length direction vector $v_k \in \mathbb{R}^D : \|v_k\|_2 = 1$, such that the parameter pair (μ_k, v_k) models a line through μ_k along the direction $\pm v_k$. To decide for membership of a datum to the k -th class, the framework/algorithm should analyze the distance from the datum to the line modeling the k -th class.

- (5 points) Mathematically formulate a hard-clustering problem as an optimization problem for the parameter set $\{(\mu_k, v_k)\}_{k=1}^K$ given the data.

- (15 points: 5 + 5 + 5) You decide to solve the optimization problem using an iterative scheme. Mathematically derive the updates to solve for the model parameters (both kinds) and the cluster labels, given the data, for solving the optimization problem formulated in the previous sub-question.

- (5 points: 3 + 1 + 1) Write clear pseudocode for an implementable algorithm, using the parameter updates derived in the previous question. Specify a suitable initialization, and a suitable termination criterion.

- (5 points) How will you modify the hard-clustering formulation of the optimization problem to a different formulation that perform soft clustering ? Clearly state this optimization problem.

2. [20 points] In the context of the EM algorithm, clearly explain the following mathematically (with suitable symbolic notation) and in words both:
- (2 points: 1 + 1) “Missing data” and “complete data”.
 - (6 points: 2 + 2 + 2) The properties of the log-likelihood function, the $F(\cdot)$ function, and the $Q(\cdot)$ function. State each function mathematically, and state all associated properties mathematically.

- (4 points) The E step and the M step, and the iterative optimization algorithm.

- (6 points: 3 + 3) Does the optimization algorithm guarantee a sequence of parameter updates with non-decreasing likelihood-function values ? Explain why, or why not ?

Does the optimization algorithm guarantee a sequence of parameter updates with non-decreasing $Q(\cdot)$ -function values ? Explain why, or why not ?

- (2 points) State a termination criterion for the optimization algorithm that can be implemented in practice.

3. [5 points] In the context of T1-weighted MR imaging, during the acquisition protocol of “saturation recovery” for a 3D brain specimen, explain the following. Assume that each voxel contains either all white matter, or all gray matter, or all cerebrospinal fluid.
- (1 points) Initial magnetization state (vector magnitude and direction), i.e., at time $t = 0$, within all voxels.
 - (2 points) Final magnetization state (vector magnitude and direction), i.e., at time $t \rightarrow \infty$, within all voxels.
 - (2 points) Intermediate magnetization state (vector magnitude and direction), i.e., at finite time $t \in (0, \infty)$, within all voxels.

4. [15 points] For fuzzy C-means clustering:
- (4 points) Formulate the optimization problem.

- (8 points: 4 + 4) Mathematically derive the updates for the cluster memberships and the cluster means.

- (3 points) Write clear pseudocode for an implementable algorithm, using the parameter updates derived in the previous question. Specify a suitable initialization, and a suitable termination criterion.